

Illegal gold mining in the Amazon: data dictionary

Data for Good, Lessons 3 to 5 · The six tables in the folder `data/` of the data pack

For all tables. Areas are in hectares (100 ha = 1 km²). **year** is the calendar year. There are no missing values: a zero means that no area was classified in that class. Every unit has a row for every year. Names are spelled as the publisher spells them, with accents (*Amazonía*, *Amazônia*, *Áncash*): read the files as UTF-8. The source of each table, with link and date of download, is in `DOWNLOAD_LOG.csv`.

1. Mining and forest area by place and year (three tables)

Source: MapBiomas land-cover statistics (Peru: Collection 4; Brazil: Collection 10.1). The three tables have these variables in common:

Variable	Definition	Unit
year	Year of the land-cover map.	year
mining_ha	Area in the land-cover class “mining” (Peru: class 4.2 <i>Minería</i> ; Brazil: class 4.3 <i>Mining</i>) in that year. All mining: legal and illegal, artisanal and industrial, gold and other minerals.	ha
forest_ha	Area in the class group “forest” (Peru: 1. <i>Formación boscosa</i> ; Brazil: 1. <i>Forest</i>). The group contains several forest classes: see the publisher’s legend.	ha
total_ha	Area in all classes together: the total area of the unit in that row.	ha

peru_department_year.csv. One row is a department in a biome in a year. 2,255 rows, 1985 to 2025. **department**, **biome** and **year** identify a row.

Variable	Definition	Values
department	Name of the department.	25 names
biome	Biome. A department can lie in more than one biome, with one row for each.	Amazonía , Andes , Desierto costero (coastal desert), Bosque seco ecuatorial (equatorial dry forest)

peru_bufferzone_year.csv. One row is the buffer zone of a protected area, in a department, in a year. 4,100 rows, 1985 to 2025. **buffer_zone**, **pa_category**, **department** and **year** identify a row. The areas refer to the buffer zone, not to the protected area inside it.

Variable	Definition	Values
buffer_zone	Name of the protected area that the buffer zone surrounds.	60 names
pa_category	Category of that protected area.	<i>Parque Nacional</i> (national park), <i>Reserva Nacional</i> , <i>Reserva Comunal</i> , <i>Santuario Nacional</i> , <i>Santuario Histórico</i> , <i>Bosque de Protección</i> , <i>Refugio de Vida Silvestre</i> , <i>Coto de Caza</i> , <i>Reserva Paisajística</i> , <i>Zona Reservada</i>
department	Department. A buffer zone can lie in more than one department, with one row for each.	22 names

There are 61 buffer zones and 60 names: **de Calipuy** is the name of both a *Reserva Nacional* and a *Santuario Nacional* (neither is in the Amazon). Identify a buffer zone by **buffer_zone** and **pa_category** together.

brazil_municipality_year.csv. One row is a municipality in a state, in a biome, in a year. 268,160 rows, 1985 to 2024. `state_acronym`, `municipality`, `biome` and `year` identify a row.

Variable	Definition	Values
<code>state_acronym</code>	Two-letter code of the state.	27 codes, for example PA (Pará), RR (Roraima), MT (Mato Grosso)
<code>state</code>	Name of the state.	27 names
<code>municipality</code>	Name of the municipality. The same name can occur in more than one state: identify a municipality by state and name together.	5,297 names
<code>biome</code>	Biome. A municipality can lie in more than one biome, with one row for each.	Amazônia, Cerrado, Caatinga, Mata Atlântica, Pantanal, Pampa

The table has 5,738 combinations of state and municipality. This is more than the number of municipalities, because a municipality on a state border can have a second row under the neighboring state, with a few hectares (see “Notes on the data” in the assignment brief).

2. Mining area in Brazil by type and substance

mining_area.csv. One row is a territory in a year. 82 rows: two territories, 1985 to 2025. Source: MapBiomias Brazil platform, Collection 11, mining theme. All areas in hectares.

Variable	Definition
<code>territory</code>	Brasil (the whole country) or all indigenous lands combined (all indigenous territories in Brazil, added up). The second is part of the first.
<code>year</code>	1985 to 2025.
<code>mining_artisanal_ha</code>	Artisanal mining (<i>garimpo</i>), all substances.
<code>mining_industrial_ha</code>	Industrial mining, all substances.
<code>gold_artisanal_ha</code>	Artisanal mining, substance recorded as gold.
<code>gold_industrial_ha</code>	Industrial mining, substance recorded as gold.
<code>nosubstance_artisanal_ha</code>	Artisanal mining for which the publisher records no substance (“No Substance”).
<code>gold_or_nosub_artisanal_ha</code>	<code>gold_artisanal_ha</code> + <code>nosubstance_artisanal_ha</code> . Our working measure of illegal gold mining (see the assignment brief).
<code>gold_share_of_artisanal</code>	<code>gold_artisanal_ha</code> / <code>mining_artisanal_ha</code> . A share between 0 and 1.
<code>artisanal_exposed,</code> <code>artisanal_reveg</code>	<code>gold_artisanal_ha</code> split in two: area with exposed soil, and area with revegetation (old mining sites where vegetation has regrown, still classified as mining). In hectares, gold only.
<code>industrial_exposed,</code> <code>industrial_reveg</code>	The same split for <code>gold_industrial_ha</code> .

Artisanal mining of other substances (for example tin) is `mining_artisanal_ha` minus `gold_or_nosub_artisanal_ha`. This table has no breakdown by state or municipality, and the three tables in section 1 have no breakdown by type or substance.

3. World commodity prices

Source: World Bank Commodity Price Data (“Pink Sheet”), September 2026, sheet “Monthly Prices”. All prices are nominal US dollars (not adjusted for inflation).

`prices_monthly.csv`. One row is a month. 799 rows, February 1960 to August 2026.

`prices_annual.csv`. One row is a year. 65 rows, 1961 to 2025. Each price is the mean of the twelve monthly prices of that year. 2026 is not included because the year is incomplete.

Variable	Definition	Unit
<code>date</code> (monthly file only)	First day of the month, written as 1960-02-01.	date
<code>year</code>	Calendar year.	year
<code>gold</code> , <code>silver</code> , <code>platinum</code>	Price of the metal.	\$ per troy ounce (31.1 grams)
<code>copper</code> , <code>tin</code>	Price of the metal.	\$ per metric ton
<code>iron_ore</code>	Price of iron ore (Pink Sheet series “Iron ore, cfr spot”).	\$ per dry metric ton unit
<code>oil</code>	Price of crude oil (Pink Sheet series “Crude oil, average”).	\$ per barrel
<code>gold_idx</code> , <code>silver_idx</code> , ... (annual file only)	Price index: the annual price divided by the price in 2018, times 100.	2018 = 100
<code>ln_gold</code> , <code>ln_silver</code> , ... (annual file only)	Natural logarithm of the annual price.	log of \$

4. Variables that are not in the tables

You construct these yourself, in your replication code:

- **Annual addition:** `mining_ha` in a year minus `mining_ha` in the year before, for the same unit. Add up the rows of a unit first (see “More than one row per place and year” in the assignment brief).
- **Exposure:** your choice, for example `mining_ha` / `total_ha` of a unit in 2010.
- **Real prices:** the tables contain nominal prices. If you want real prices, you need a price index from another source, which you log and cite.

The tables do not contain: whether a mining site is legal; mining by type or substance for Peru or for Brazilian municipalities; the area inside protected areas or inside individual indigenous territories. Some of these are in the table that your question card asks you to add.